

Q-2009, B
Question: Define numerical integration and obtain Trapezoidal rule for numerical integration and General Quadrature formula.

Sol: The process of computing the value of a definite integral from a set of numerical values of the integrand is called numerical integration.

Let $I = \int_a^b y \, dx$ where $y = f(x)$. Let $f(x)$ be given for certain equidistance values of x say $x_0, x_0+h, x_0+2h, \dots$. Let the range (a, b) be divided into n equal parts, each of width h . So that $b-a = nh$.

Let $x_0 = a, x_1 = x_0 + h = a+h, x_2 = a+2h, \dots, x_n = a+nh = b$. We have assume that the $(n+1)$ ordinates $y_0, y_1, \dots, y_n$ are at equal intervals.

$$I = \int_a^b y \, dx = \int_{x_0}^{x_0+nh} y \, dx = \int_0^n y_{x_0+uh} h \, du, \quad \text{where } u = \frac{x-x_0}{h} \Rightarrow dx = h \, du$$

$$\text{or, } \frac{1}{h} \int_0^n y \, du$$

$$\text{or, } I = h \int_0^n \left[y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots \right] du$$

$$\Rightarrow I = \int_{x_0}^{x_0+nh} y \, dx = h \left[y_0 + \frac{y_1 - y_0}{2} + \dots \right]$$

$$\begin{aligned}
 \Rightarrow I &= \int_{x_0}^{x_0+nh} y \, dx = h \int_0^n \left[y_0 + u \Delta y_0 + \frac{(u^2-u)}{2} \Delta^2 y_0 + \frac{(u^3-3u^2+2u)}{6} \Delta^3 y_0 + \frac{(u^4-6u^3+11u^2-6u)}{24} \Delta^4 y_0 + \dots \right] du \\
 &= h \int_0^n \left[y_0 + u \Delta y_0 + \frac{1}{2}(u^2-u) \Delta^2 y_0 + \frac{(u^3-3u^2+2u)}{6} \Delta^3 y_0 + \frac{(u^4-6u^3+11u^2-6u)}{24} \Delta^4 y_0 + \dots \right] du \\
 &= h \int_0^n \left[y_0 + u \Delta y_0 + \frac{1}{2}(u^2-u) \Delta^2 y_0 + \frac{(u^3-3u^2+2u)}{6} \Delta^3 y_0 + \frac{(u^4-6u^3+11u^2-6u)}{24} \Delta^4 y_0 + \dots \right] du \\
 &= h \left[u y_0 + \frac{u^2}{2} \Delta y_0 + \frac{1}{6} \left(\frac{u^3}{3} - \frac{3u^2}{2} + \frac{2u}{2} \right) \Delta^2 y_0 + \frac{1}{24} \left(\frac{u^4}{4} - \frac{6u^3}{3} + \frac{11u^2}{2} - \frac{6u}{2} \right) \Delta^3 y_0 + \dots \right]_0^n \\
 &= h \left[n y_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{6} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2!} + \left(\frac{n^4}{24} - \frac{n^3}{6} + \frac{n}{2} \right) \frac{\Delta^3 y_0}{3!} + \dots \right] \text{ up to } (n+1) \text{ terms}
 \end{aligned}$$

This is the general quadrature formula.

putting $n=1$ in ① and neglecting second and higher order difference, we get

$$\int_{x_0}^{x_0+h} y \, dx = h \int_{x_0}^{x_0+h} \left(y_0 + u \Delta y_0 \right) dx$$

$$\begin{aligned}
 &= h \left[y_0 + \frac{1}{2} \Delta y_0 \right] \\
 &= h \left[y_0 + \frac{y_1 - y_0}{2} \right] = h \left[\frac{y_0 + y_1}{2} \right] \\
 &= h \left[\frac{y_0 + y_1}{2} \right]
 \end{aligned}$$

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 &= h \left[\frac{y_0 + y_1}{2} \right] = h \left[\frac{y_0 + y_1}{2} \right] \\
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 \end{aligned}$$

distance between two consecutive ordinates

$\bar{x}$ means of the first and the last ordinates + sum of all the intermediate ordinates

This rule is known as the Trapezoidal Rule.

Question: Derive $\frac{1}{3}$ Simpson's Rule for Numerical integration.

Soln Let $I = \int_a^b y dx$ where $y = f(x)$. Let $f(x)$ be given for certain equidistance values of x say $x_0, x_0+h, x_0+2h, \dots$ Let the range (a, b) be divided into n equal parts, each of width h .

so that $b-a = nh$

Let $x_0 = a, x_1 = x_0 + h = a + h, x_2 = a + 2h, \dots, x_n = a + nh = b$

We have assumed that the $(n+1)$ ordinates are at equal intervals.

$$I = \int_a^b y dx = \int_{x_0}^{x_0+nh} y dx = \int_{x_0}^{x_0+nh} y \frac{dx}{dx} = \int_{x_0}^{x_0+nh} y \frac{1}{h} du$$

where, $u = \frac{x-x_0}{h}$

or, $I = h \int_0^n [y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots]$

$$= h \int_0^n [y_0 + u \Delta y_0 + \frac{u^2 u}{2!} \Delta^2 y_0 + \frac{1}{3!} (u^3 u) \Delta^3 y_0 + \dots] du$$

$$= h \int_0^n [y_0 + u \Delta y_0 + \frac{u^3 u}{2!} \Delta^2 y_0 + \frac{1}{3!} (u^3 2u - u^4 + 2u) \Delta^3 y_0 + \dots] du$$

$$= h \int_0^n [y_0 + u \Delta y_0 + \frac{1}{2!} (u^3 u) \Delta^2 y_0 + \frac{1}{3!} (u^3 3u^2 + 2u) \Delta^3 y_0 + \dots] du$$

$$\therefore I = h [y_0 + \frac{1}{2} \Delta y_0 + \frac{1}{2!} (\frac{u^3 - 0^3}{3} \Delta^2 y_0 + \frac{1}{3!} (u^3 \frac{u^2}{2} - \frac{3u^3}{3} + \frac{2u^3}{3}) \Delta^3 y_0 + \dots]$$

$$= h [y_0 + \frac{1}{2} \Delta y_0 + \frac{1}{2!} (\frac{u^3}{3} - \frac{u^3}{3}) \Delta^2 y_0 + \frac{1}{3!} (\frac{u^4}{4} - \frac{u^3}{3} + \frac{2u^3}{3}) \Delta^3 y_0 + \dots]$$

This is general Quadrature formula

Putting $n=2$ in equation (1) and neglecting third and higher order differences we get

$$\int_{x_0}^{x_0+2h} y dx = h [2y_0 + 2 \Delta y_0 + \frac{1}{3} (\frac{2^3}{3} - \frac{2^3}{3}) \Delta^3 y_0 + \dots]$$

$$\int_{x_0}^{x_0+h} y dx = h \left[y_0 + 2y_1 + \frac{1}{2} \left(\frac{8}{3} y_2 + 2 \right) \Delta y_0^2 \right] \quad n=1$$

$$= h \left[y_0 + 2y_1 + \frac{1}{2} \left(\frac{8}{3} y_2 + 2 \right) \Delta y_0^2 \right]$$

$$= h \left[y_0 + 2y_1 + \frac{1}{2} \left(\frac{8}{3} y_2 + 2 \right) \Delta y_0^2 \right]$$

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$$= h \left[y_0 + 2y_1 + \frac{1}{2} \left(\frac{8}{3} y_2 + 2 \right) \Delta y_0^2 \right]$$

$$= \frac{1}{3} \left(y_0 + y_1 + y_2 \right) h \quad n=1$$

$$= \frac{1}{3} \left(y_0 + y_1 + y_2 \right) h$$

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$$= \frac{1}{3} \left(y_0 + y_1 + y_2 \right) h$$

Adding all these integrals, we get

$$\int_{x_0}^{x_0+h} y dx = \frac{1}{3} \left[(y_0 + y_1) + 4(y_1 + y_2) + \dots + (y_{n-1} + y_n) \right]$$

This formula is known as Simpson's $\frac{1}{3}$ Rule.

Problem: Find $\int_0^1 x dx$ by the trapezoidal rule and by Simpson's $\frac{1}{3}$ rule. when $n=8$.

Solⁿ: Here, $n=8$

Here, $n=8$

$$h = \frac{b-a}{n} = \frac{1-0}{8} = 0.125$$

Let $y = \frac{1}{3}x^2$ for each point of subdivision. The function $y = \frac{1}{3}x^2$ is shown in the following table.

Calculated values are as shown in the following table.

$x_0 = 0$	$y_0 = 0$
$x_1 = 0.125$	$y_1 = 0.0088$
$x_2 = 0.25$	$y_2 = 0.0080$
$x_3 = 0.375$	$y_3 = 0.00727$
$x_4 = 0.500$	$y_4 = 0.00667$
$x_5 = 0.625$	$y_5 = 0.00615$
$x_6 = 0.750$	$y_6 = 0.00571$
$x_7 = 0.875$	$y_7 = 0.00533$
$x_8 = 1$	$y_8 = 0.00500$

(a) By Trapezoidal rule, we have

$$\begin{aligned}\int_0^1 \frac{1}{1+x} dx &= \frac{h}{2} [y_0 + y_8 + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7)] \\ &= \frac{0.125}{2} [1 + 0.5 + 2(0.888 + 0.8 + 0.727 + 0.667 + 0.615 + 0.571 + 0.533)] \\ &= \frac{0.125}{2} [1.5 + 9.602] \\ &= \frac{0.125 \times 11.102}{2} \\ &= 0.693\end{aligned}$$

(b) By Simpson's $\frac{1}{3}$ rule, we have

$$\begin{aligned}\int_0^1 \frac{1}{1+x} dx &= \frac{h}{3} [y_0 + y_8 + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6)] \\ &= \frac{0.125}{3} [1 + 0.5 + 4(0.888 + 0.727 + 0.615 + 0.533) + 2(0.8 + 0.667 + 0.571)] \\ &= \frac{0.125}{3} (1.5 + 11.052 + 4.076) \\ &= \frac{0.125 \times 16.628}{3} \\ &= 0.693\end{aligned}$$

Ans:

1	0.888
0.888	0.8
0.8	0.727
0.727	0.667
0.667	0.615
0.615	0.571
0.571	0.533
0.533	0.5
0.5	0.4
0.4	0.333
0.333	0.286
0.286	0.25
0.25	0.222
0.222	0.2
0.2	0.182
0.182	0.167
0.167	0.154
0.154	0.143
0.143	0.133
0.133	0.125
0.125	0.118
0.118	0.111
0.111	0.105
0.105	0.1